

NIKHIL MAHADIK

Mumbai Maharashtra

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Objective

Backend & AI Software Engineer with 4+ years of experience building **production-grade** FastAPI microservices, LLM agent integrations, and **RAG pipelines**. Expert in **PostgreSQL**, **async patterns**, **REST API design**, and **enterprise system integration**. Proficient in **OpenAI/Anthropic APIs** at scale.

Experience

Infosys Limited | Backend & AI Engineer

July 2025-Present

- **Developed FastAPI microservices for GenAI at Infosys Limited, integrating RAG pipelines with OpenAI/Anthropic APIs and managing async background tasks, token limits, and streaming responses to optimize high-throughput AI tool endpoints.**
- **Developed FastAPI microservices integrating LLMs via OpenAI API, managing streaming responses and stateful workflows using LangGraph and PostgreSQL**
- **Developed FastAPI microservices integrating OpenAI LLMs with Qdrant vector stores to build RAG pipelines, optimizing asynchronous processing and retrieval precision by ~30% for AI agents.**
- **Developed FastAPI microservices to integrate LLMs, managing token streams and implementing Pydantic output parsers to ensure schema compliance and reliable API data exchange.**
- **Developed FastAPI microservices and integrated LLM APIs with streaming; designed PostgreSQL schemas for RAG pipelines; deployed containerized services via Docker and GitHub Actions, while ensuring secure authentication and high-throughput async API performance at Infosys Limited.**

Projects

AI HR Policy API & RAG Tooling | FastAPI, LLM Integration, PostgreSQL, Docker

Jan 2025

- **Developed FastAPI microservices for RAG agents at Infosys Limited, automating policy query resolution and redaction.**
- **Developed scalable FastAPI microservices → integrating OpenAI LLMs → managed async streaming, token limits, and RAG pipelines using PostgreSQL and Qdrant for efficient retrieval.**
- **Developed FastAPI microservices exposing RAG capabilities via REST APIs, integrating OpenAI/Anthropic LLMs with → async background workers and PostgreSQL-backed**
- **Developed FastAPI microservices with Docker; integrated LLM APIs with async patterns to manage streaming, token limits, and complex RAG workflow optimizations reducing infrastructure costs by 25% through strategic caching and PostgreSQL query tuning.**
- **Developed FastAPI microservices using async patterns to achieve 800ms P95 latency integrating OpenAI/Anthropic LLM APIs with optimized prompt streaming and RAG.**

Skills

Backend & API: FastAPI, Python, Asyncio, RESTful Microservices, Request Validation, Middleware, Dependency Injection

AI Integration: OpenAI/Anthropic APIs, Streaming, Token Management, RAG Pipelines, Tooling, LangGraph, Guardrails

Data & Systems: PostgreSQL, MongoDB, SQL Query Optimization, Indexing, SharePoint & ERP Integrations, Vector DBs

Engineering Ops: Docker, Git, CI/CD, Background Tasks, Queueing, AuthN/AuthZ, High-Throughput APIs

Observability: LangSmith (Tracing, Evaluation), Latency Monitoring, Error Handling, Logging, Unit Testing

Languages: Python (Production), SQL, Java, C++, JavaScript

Achievements

- **Leetcode (Max Rating 1797)** | Top 5% in India Among 100,000 active users.  [Leetcode](#)
- **3 Star - Codechef (Max Rating 1628)** | Top 5% Among 170K+ Active Users  [Codechef](#)
- **Pupil - Codeforces (Max Rating 1206)** | Top 10% Among 70K+ Active Users  [Codeforces](#)
- **1st Place in VNPS** | National Level Project Showcase
- **3rd place in CodeCraze** | Data Structures and Algorithm Implementation Based Competition

Education

Vidyavardhini's College of Engineering And Technology

May 2025

Bachelor of Engineering in Computer Science - **CGPA - 8.5**